

Representations of the Fundamental Group and Parabolic Bundles

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The study of the association between holomorphic vector bundles on a compact Riemann surface (equivalently, a smooth algebraic curve) and representations of its fundamental group dates back to the pioneering work of A. Weil in the 1930s. Weil's seminal article, where he classifies vector bundles that arise from representations, inspired Narasimhan and Seshadri in their quest to construct moduli spaces of vector bundles on compact Riemann surfaces. Around the same time, in a different part of the world, Mumford introduced a general recipe for constructing moduli spaces using geometric invariant theory. The celebrated Narasimhan-Seshadri theorem beautifully unites these approaches, asserting that both constructions yield the same moduli space. Later, Mehta and Seshadri extended these ideas to punctured compact Riemann surfaces, introducing the concept of parabolic bundles. This work opened new avenues for understanding vector bundles on curves with marked points. While much is now understood about vector bundles and parabolic bundles on smooth curves, analogous studies on singular curves have progressed more slowly. Singularities, such as nodes, introduce challenges that require careful treatment. In this talk, we focus on irreducible nodal curves and explore the question: Is the Mehta-Seshadri theorem true for nodal curves? This is joint work with Dr. Sanjay Singh. I will begin by setting up the necessary background and defining the key objects, making the talk accessible and self-contained. Following this, I will discuss our contributions toward answering the above question.